

Smart Operator Assistant for CAT machinery

Background:

Construction equipment like excavators and loaders are becoming increasingly digitalized, yet the tools available to machine operators remain basic. Imagine an intelligent assistant – an – end to end application – that supports machine operators throughout their workday, improving efficiency, safety and training using smart technologies.

Challenge:

Design and build a multi-functional operator interface for CAT machines operators. Participants are encouraged to think beyond just a tool – make it an intelligent companion that enhances the operator's daily experience.

Expected outcomes:

- **Daily task dashboard:** View scheduled tasks for the day.
- **Safety Features:** Implement features that improve real-time operator safety using available or assumed data.
 1. Seatbelt compliance,
 2. proximity hazards,
 3. incident logging etc.(Working conditions to be considered)
- **Operator training hub:** Choose any creative learning format – e-learning videos, instructor booking or simulation module.
- **Identify unusual behavior in machine usage e.g. excessive idling or unsafe operation patterns**
- **Task Time Estimation:** Predict time to complete a task based on past data and environmental conditions

Timestamp	Machine ID	Operator ID	Engine Hours	Fuel Used (L)	Load Cycles	Idling Time (min)	Seatbelt Status	Safety Alert Triggered
2025-05-01 08:00:00	EXC001	OP1001	1523.5	5.2	12	30	Fastened	No
2025-05-01 10:00:00	EXC001	OP1001	1524.8	3.8	2	55	Unfastened	Yes
2025-05-01 14:00:00	EXC001	OP1001	1526.5	6.1	10	15	Fastened	No
2025-05-02 09:00:00	EXC001	OP1001	1530.2	2.0	1	60	Unfastened	Yes

Task ID	Task Type	Weather	Operator Skill	Machine Age (yrs)	Estimated Time (min)	Actual Time (min)
T001	Earth Excavation	Sunny	Expert	2	60	58
T002	Trenching	Rainy	Intermediate	4	45	52
T003	Material Loading	Cloudy	Beginner	3	30	42
T004	Grading	Sunny	Expert	5	35	33
T005	Demolition	Windy	Intermediate	6	90	105